Foundations of ADCS

5. ADCS Simulation

Rishav
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Foundations of ADCS

https://github.com/risherlock/ADCS

0. Course introduction
1. Preliminaries of ADCS
2. Attitude determination system
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5. **ADCS simulation**

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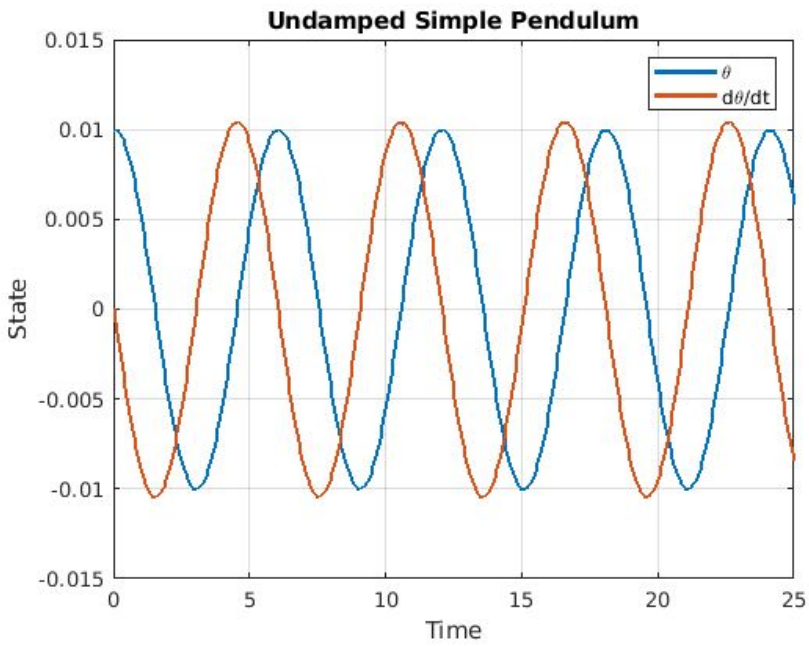
simulation

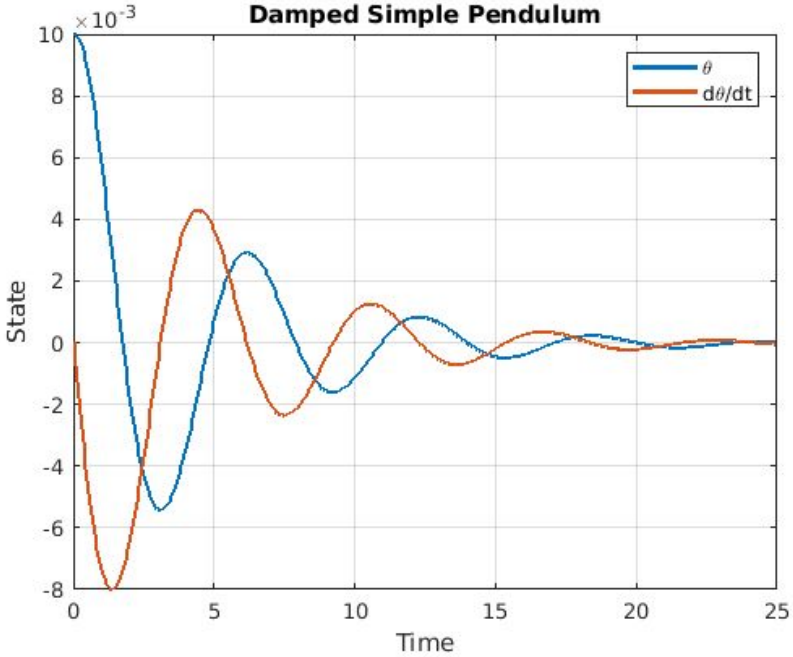
(noun) a situation in which a particular set of conditions is created artificially in order to study or experience something that could exist in reality.

Why simulation?

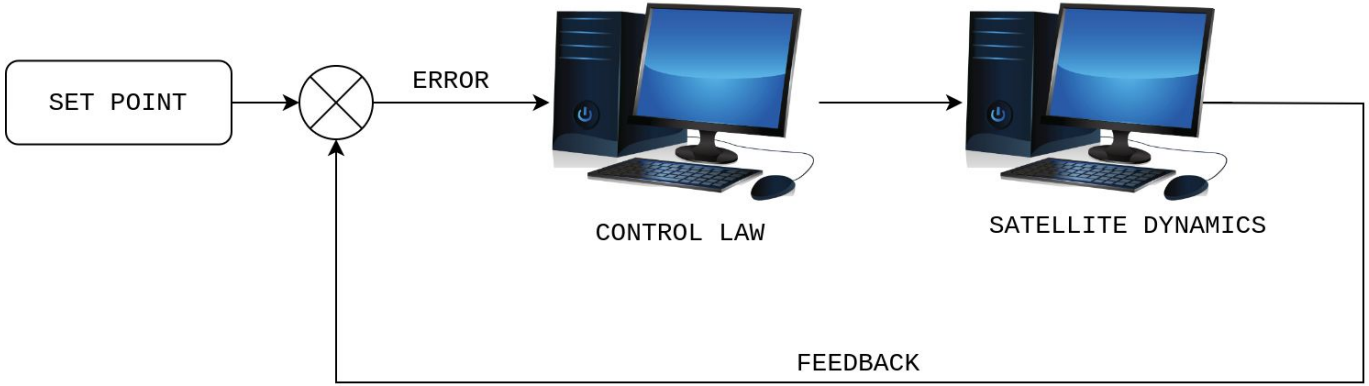
1. Space is not accessible 24/7
2. Reduces time of R&D
3. Proof of concepts, designs, and algorithms
4. Prediction of spacecraft behavior
5. Control gain tuning and sensitivity analysis
6. Reduces development cost
7. Reduces risks involved (both human and expensive devices)

Expectations of simulation

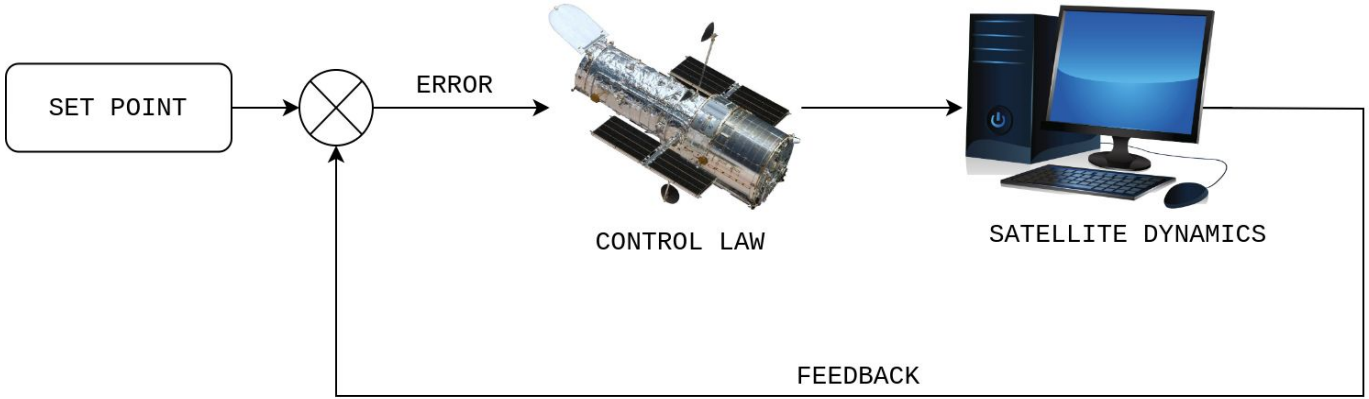




SIL simulator



HIL simulator



How to perform numerical simulation of dynamical systems?

Differential equations

Newton's second law: $\boldsymbol{F} = m\dfrac{d^2x}{dt^2}$

Euler's equations: $\boldsymbol{I}\dot{\boldsymbol{\omega}} + \omega \times (\boldsymbol{I}\boldsymbol{\omega}) = \boldsymbol{\tau}$

Simple pendulum: $\dfrac{d^2\theta}{dt^2} + \dfrac{b}{m}\dfrac{d\theta}{dt} + \dfrac{g}{L}\sin\theta = 0$

Schrödinger equation: $\dfrac{1}{c^2}\dfrac{\partial^2\phi_n}{\partial t^2} - \nabla^2\phi_n + (\dfrac{mc}{\boldsymbol{\hbar}})^2\phi_n = 0$

Generic representation

$$\dot{\boldsymbol{x}} = \boldsymbol{f}(\boldsymbol{x}) \qquad \boldsymbol{x} \in \mathbb{R}^n$$

Autonomous differential equation

$$\dot{\boldsymbol{x}} = \boldsymbol{f}(\boldsymbol{x}, t) \qquad \boldsymbol{x} \in \mathbb{R}^n$$

Non-autonomous differential equation

Conversion to 1st order form

$$\ddot{\theta} + \frac{d}{m}\dot{\theta} + \frac{g}{L}\sin\theta = 0$$

Simple pendulum

$$\boldsymbol{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} \theta \\ \dot{\theta} \end{bmatrix}$$

State vector

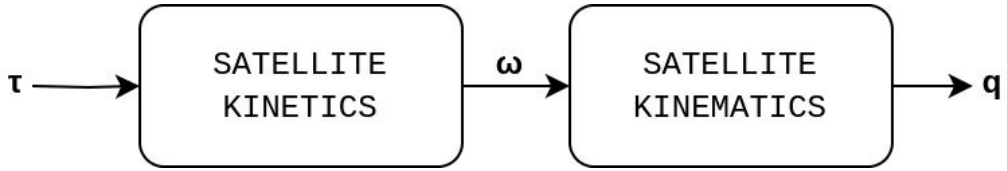
$$\dot{\boldsymbol{x}} = \begin{bmatrix} x_2 \\ -\frac{d}{m}x_2 - \frac{g}{L}\sin x_1 \end{bmatrix}$$

State space representation

$$\dot{\boldsymbol{x}} = \boldsymbol{f}(\boldsymbol{x}) \qquad \boldsymbol{x} \in \mathbb{R}^2$$

First-order form

Satellite dynamics



Control torques and external disturbances $\boldsymbol{\tau}$ change the angular velocity $\boldsymbol{\omega}$ of satellite which, in turn, changes the attitude $\mathbf{q}$.

Kinetics

$$\boldsymbol{I}\dot{\boldsymbol{\omega}} = -\boldsymbol{\omega} \times (\boldsymbol{I}\boldsymbol{\omega}) + \boldsymbol{\tau}$$

where, $\boldsymbol{I} \in \mathbb{R}^{3\times 3}$ is moment of inertia matrix, $\boldsymbol{\omega} \in \mathbb{R}^3$ is angular velocity, and $\boldsymbol{\tau} \in \mathbb{R}^3$ is external torque to the satellite.

Kinematics

$$\dot{\boldsymbol{q}} = 0.5 \begin{bmatrix} 0 & -\omega_1 & -\omega_2 & -\omega_3 \\ \omega_1 & 0 & \omega_3 & -\omega_2 \\ \omega_2 & -\omega_3 & 0 & \omega_1 \\ \omega_3 & \omega_2 & -\omega_1 & 0 \end{bmatrix} \boldsymbol{q}$$

$$\boldsymbol{q} = [q_0,\, q_1,\, q_2,\, q_3]^\intercal \in \mathbb{R}^4$$

Numerical integration

$$\boldsymbol{x}_{\text{new}} = \boldsymbol{x}_{\text{old}} + \dot{\boldsymbol{x}}\, dt$$

Elements of numerical integration:

1. Initial state
2. Differential equation
3. Physical parameters (eg. moment of inertia)
4. Time step (dt)
5. Simulation time

Examples

1. Simple pendulum
2. Satellite dynamics
3. Satellite rate control
4. Satellite attitude control

Exercise: Attitude control law

$$\boldsymbol{\tau} = -k_p \,\mathrm{sign}(\delta q_0) \begin{bmatrix} \delta q_1 \\ \delta q_2 \\ \delta q_3 \end{bmatrix} - k_d \begin{bmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \end{bmatrix}$$

where, k_p and k_d are positive scalar feedback gain for attitude and angular rates respectively.

Source:

Eqn.(7.12) from Markley, Crassidis - *Fundamentals of Spacecraft Attitude Determination and Control* (2014)

Thank you!